XIAO JIN

Senior Data Scientist · Applied Scientist · Pricing, Revenue Management & Optimization

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SUMMARY

Ph.D. operations researcher and data scientist with six years building decision systems that reach production. I lead projects end to end — problem framing with commercial stakeholders, modelling, engineering, deployment and post-launch measurement — across airline pricing and revenue management, healthcare operations and logistics. Deep-learning choice models, simulation–optimization and full-stack delivery with LLM coding agents (Python / PyTorch / Django / React). Currently lead researcher for a corporate-lab portfolio with airline partners.

IMPACT HIGHLIGHTS

- **7–8% lift in checked-baggage revenue ($p < 0.01$)** — measured in a **randomized field deployment** against a uniform-price control (regression-adjusted); developed the method and ran the field implementation, now **running in production at a partner airline**. Average posted price was lower than under the incumbent rule. First author on the resulting paper, under review at *Operations Research*.
- **Call abandonment under the 5% target (about 4%)** — simulation–optimization rostering model in a **2024 field pilot** at the NUHS JMC call centre.
- **∼18–21% projected cost saving** on SKU repositioning — simulation and data-driven forecasting backbone for Kuehne+Nagel inventory operations.
- **Transformer-based deep choice model in production** — trained on customer browsing and transactional data to predict ticket purchase; embedded in the booking simulator and used by the airline's practitioners to stress-test pricing and RM decisions before deployment.

PROFESSIONAL EXPERIENCE

SIA–NUS Digital Aviation Corporate Laboratory — Senior Research Fellow *Nov 2024 – Present*

Joint corporate R&D lab of **Singapore Airlines Group** and NUS — lab research runs in production at a group carrier. Research Fellow, same lab, Nov 2022 – Nov 2024 · Singapore

- **Lead researcher, Phase 1 portfolio** (Revenue Management & Network Planning) — own the research agenda and delivery across workstreams; coordinate milestones and stakeholder reporting; onboard and train new team members.
- **Checked-baggage pricing, partner airline (project lead, shipped).** Built a rational-inattention demand model, trained on 1.41M booking sessions across five baggage tiers, that prices the whole tier ladder from a handful of interpretable parameters — calibrated by LP so it scales to thousands of sparse segments. **Randomized field deployment (59,141 sessions): 7–8% baggage revenue lift vs a uniform-price control (regression-adjusted), $p < 0.01$**, with a lower average posted price than the incumbent rule. In production.
- **Airline revenue-prediction platform (lead researcher, adopted).** End-to-end simulation platform forecasting revenue over 1-week to 1-year selling horizons under competitor pricing, RM policy and network-level capacity competition. Trained and deployed a **Transformer-based deep choice model** on browsing and transaction data; built the full stack (**Django + React, MongoDB, Celery**) with agentic AI-assisted development (Claude Code). Used by the airline's practitioners to validate pricing and RM policy pre-deployment.
- **LLM auto-formulation paper (case study).** Contributed the airline revenue-management case study to a multi-agent LLM auto-formulation paper (7th of 10 authors; *Operations Research*, major revision; arXiv:2601.09635).
- **Call-centre rostering, NUHS JMC (project lead).** Discrete-event simulation plus optimization for agent rostering; **kept abandonment under the 5% target (about 4%) in a 2024 field pilot**.

- **Appointment scheduling, National Heart Centre Singapore (project lead).** Scheduling optimization under demand uncertainty to hold the centre's lead-day service promise.

Centre for Next Generation Logistics (C4NGL), NUS — Research Fellow *Oct 2020 – Nov 2022*

- **Tuas Port Finger 1 capacity study (project lead).** Full-scale discrete-event simulation — realistic ship arrivals, container profile distributions, yard layout, crane and AGV dynamics — to estimate annual TEU throughput under candidate layouts. Findings endorsed by the project stakeholder and used to inform layout decisions.
- **Kuehne+Nagel SKU repositioning (core researcher).** Simulation and simulation-based optimization backbone with data-driven demand forecasting for multi-warehouse inventory repositioning; **~18–21% projected cost reduction**.

C4NGP, NUS — Research Engineer *Aug 2019 – Oct 2020*

- Built the simulation-analytics and contextual-optimization methods that underpin the doctoral thesis and the later industrial deployments — budget-allocated experimentation under noisy, context-dependent outcomes.

DB Schenker, Asia Pacific Regional Office — Intern, Innovation Team *Aug 2017 – Nov 2017*

- Scoped and proposed the scenario-based simulation-optimization framework for inventory modelling that became the core of the doctoral thesis.

TECHNICAL SKILLS

Languages — Python (primary), SQL, C#, C, MATLAB, LaTeX

ML / AI — PyTorch, Transformers, deep discrete-choice models, scikit-learn, XGBoost, pandas, NumPy; forecasting; LLM coding agents for production delivery (Claude Code); design and causal analysis of randomized pricing experiments

Optimization & simulation — Gurobi, CPLEX; stochastic and simulation-based optimization; discrete-event simulation (custom C# / C++ engines, SimPy); ranking & selection

Engineering — Django, React, MongoDB, Celery, Git, Docker, AWS / GCP

Domains — airline revenue management, ancillary and dynamic pricing, assortment optimization, service operations, port and warehouse logistics

EDUCATION

National University of Singapore — Ph.D. *Aug 2015 – Jun 2020*

Ph.D., Industrial Systems Engineering & Management — simulation optimization for context-dependent decisions under noisy, expensive evaluation. GPA 4.21 / 5.00

Nanjing University — B.A., School of Management and Engineering *Sep 2011 – Jun 2015*

GPA 85 / 100

SELECTED PUBLICATIONS AND TALKS

- **First author**, "Multi-Product Pricing under Rational Inattention: Evidence, Estimation, and Field Deployment from Airline Ancillaries." *Under review, Operations Research* — the method deployed at the partner airline.
- Jin, X., Shen, Y., & Lee, L. H. "Contextual Optimizer through Neighborhood Estimation for Prescriptive Analytics." *INFORMS Journal on Computing*, 2026.
- Co-author, "Large-Scale Optimization Model Auto-Formulation: Harnessing LLM Flexibility via Structured Workflow" (*Operations Research*, major revision; arXiv:2601.09635).
- "Robust Ancillary Pricing for Airlines: When a Single Mark-up Outperforms Personalisation." INFORMS Annual Meeting, 2025. Invited speaker at **Sabre** on simulation methodology for RM practitioners.
- Ten peer-reviewed conference papers on simulation optimization (Winter Simulation Conference; IEEE CASE, 2017–2022) — full list: scholar.google.com/citations?user=NffLyjoAAAAJ